ECS 174 Proposal

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Name:
Infinity Edge - (Monocular Depth Estimation)

Problem:
Humans perceive how far away objects are using visual cues. However, robots lack this ability so this project aims to help them better understand depth and distance of their surroundings using depth maps.

Motivation:
Autonomous robots are constantly being advanced to perform basic human functions such as picking up objects, but how do they determine the objects' relative distance from themselves? We aim to answer this question by training a model, that when given a 2D RGB image, it will generate an accurate depth map. If this problem is solved in upcoming years, this can enhance the capabilities of monocular cameras and help reduce costs, pivoting from reliance on expensive sensors such as LiDAR.

Dataset:
https://www.kaggle.com/datasets/thanhbnhphan/sun-rgbd-2d

Method:
The planned architecture consists of a U-Net (with skip connections) decoder with the pretrained model ResNet-34 (provided by PyTorch) as the encoder. The choice of using the pretrained model may be necessary as it will help to improve accuracy and overall computation time via less epochs (hypothetically). The loss function we will use is a combination of a Masked Smooth L1 Loss and a Depth Gradient Loss. Using Google Colab L4 GPU for training, we expect the total train time to range from anywhere from a couple of hours to a day.

The evaluation metrics we will use to gauge the accuracy of our model is Mean Absolute Error, Root Mean Squared Error, Absolute Relative Error, and Threshold Accuracy. We will compare it with open-source models like MiDaS, to compare our model's effectiveness.

Final Outcome:
The user will be able to provide an image as input and the model will produce a prediction depth map of that image.